DDA

Contest Tracker · Power BI Portfolio Project

DDA Contest Performance & Student Intelligence Dashboard

Analyzing Contest Performance, Rankings & Programming Skills using Power BI

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Power BI | SQL | Excel

INTRODUCTION

Project Overview

The DDA Contest Tracker is a Power BI analytics project built to understand how students perform across coding and competitive contests. It brings together contest records, academic details, student master data, and leaderboard results into a single connected model — turning scattered spreadsheets into a clear, interactive picture of participation, accuracy, and skill.

From raw contest logs to performance intelligence

Four source tables are modeled into a star schema, then explored through two dashboards covering executive-level contest metrics and individual student performance analytics.

4 Source Tables

Contest, students, academics & leaderboard data

Star-Schema Model

Connected via contest_id & student_id

2 Interactive Dashboards

Executive overview & performance intelligence

GOALS

Project Objective

Track Participation

Monitor student participation across every contest type and cycle.

Analyze Performance

Study scores, accuracy, and overall contest rankings.

Evaluate Skill Areas

Assess MCQ accuracy and programming performance separately.

Compare Branches

Benchmark participation and performance across branches & colleges.

Classify Performers

Identify high, medium, and low performing student segments.

Track Individuals

Follow individual student performance over time.

Enable Mentoring

Support data-driven mentoring and training decisions.

THE CHALLENGE

Problem Statement

Contest and leaderboard data naturally grows across many disconnected sources. Reviewed manually, it becomes very difficult to extract consistent, timely insight.

Data scattered across multiple tables

Contest, student, and leaderboard records live separately.

Hard to track individual performance

No unified view of a single student's journey over time.

Manual ranking & accuracy analysis

Calculating accuracy and rank trends by hand is slow and error-prone.

Difficult branch/college comparison

No easy way to benchmark participation across branches.

Hard to spot high/low performers

Consistent performers are difficult to identify without a dashboard.

Power BI as the Solution

One connected model, interactive dashboards, instant insight.

DATA SOURCES

Dataset Overview

DIMENSION

dim_contest

Holds contest-level information — what each contest is, its type, and when it ran.

COLUMNS

contest_date · contest_name · contest_id · contest_type

DIMENSION

dim_student_year

Holds academic / student-year information tied to each student.

COLUMNS

branch · course_duration · current_year · email · passout_year · student_id

DIMENSION

dim_students

Master record of student identity — name, branch, and college.

COLUMNS

branch · college · email · name · passout_year · student_id · total_students

FACT TABLE

fact_leaderboard

Contest-level student performance & leaderboard metrics — the analytical core.

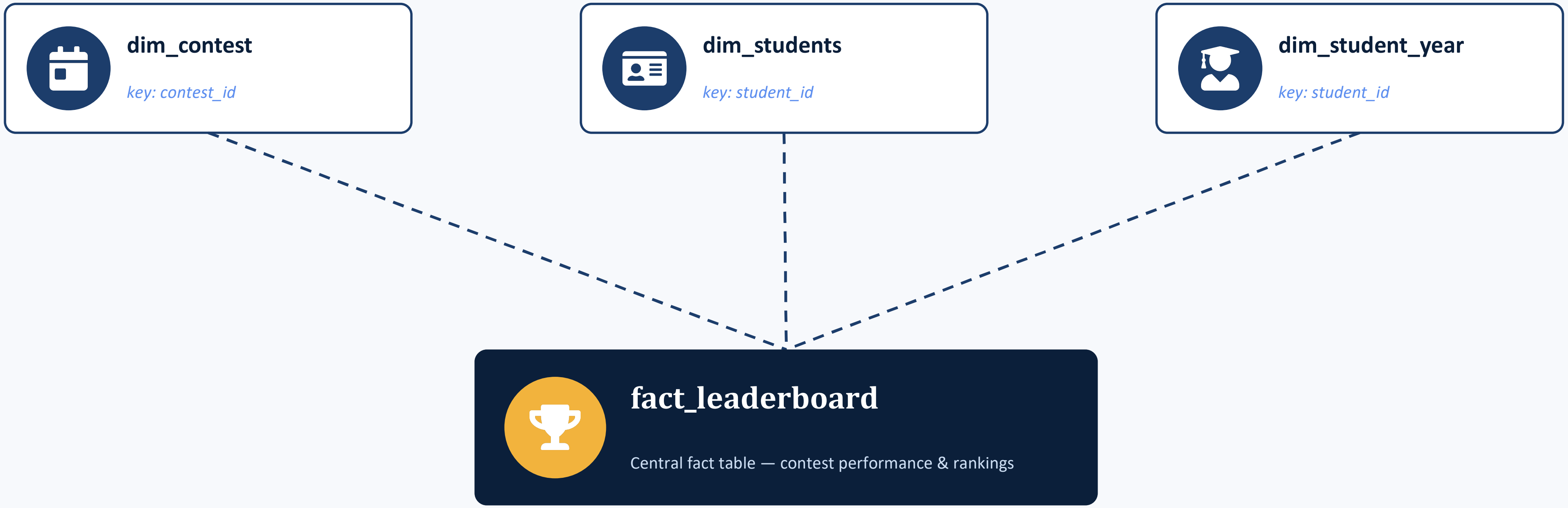
COLUMNS

accuracy · MCQ_correct / total · programming_point · rank · total_points · performance_type ...

DATA MODEL

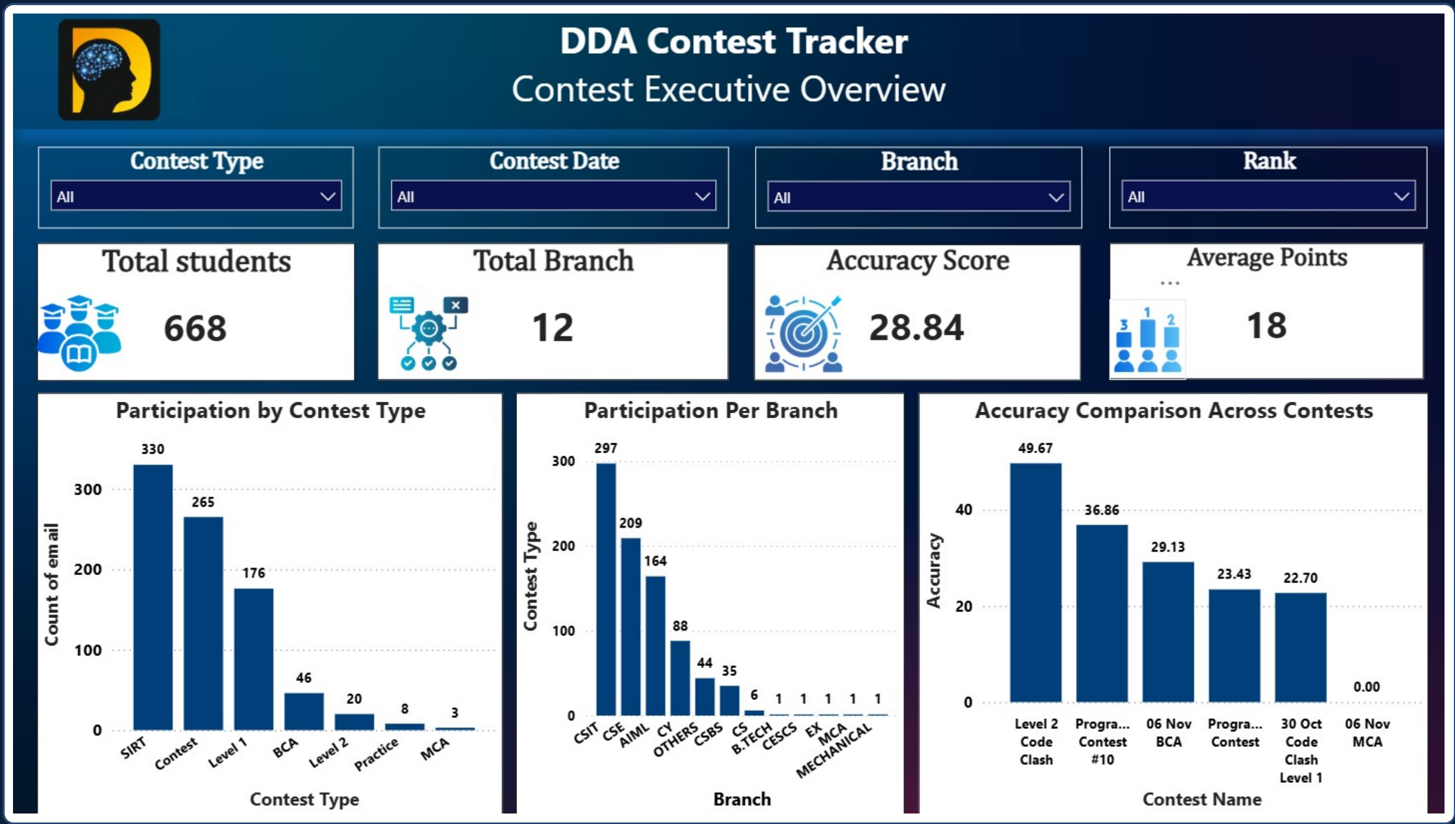
Star-Schema Design

The model follows a classic star schema — three dimension tables surround a single fact table, connected by shared key columns.



Relationships: dim_contest → fact_leaderboard (contest_id)   dim_students → fact_leaderboard (student_id)   dim_student_year → fact_leaderboard (student_id)

Contest Performance — Executive Overview



WHAT IT ANALYZES

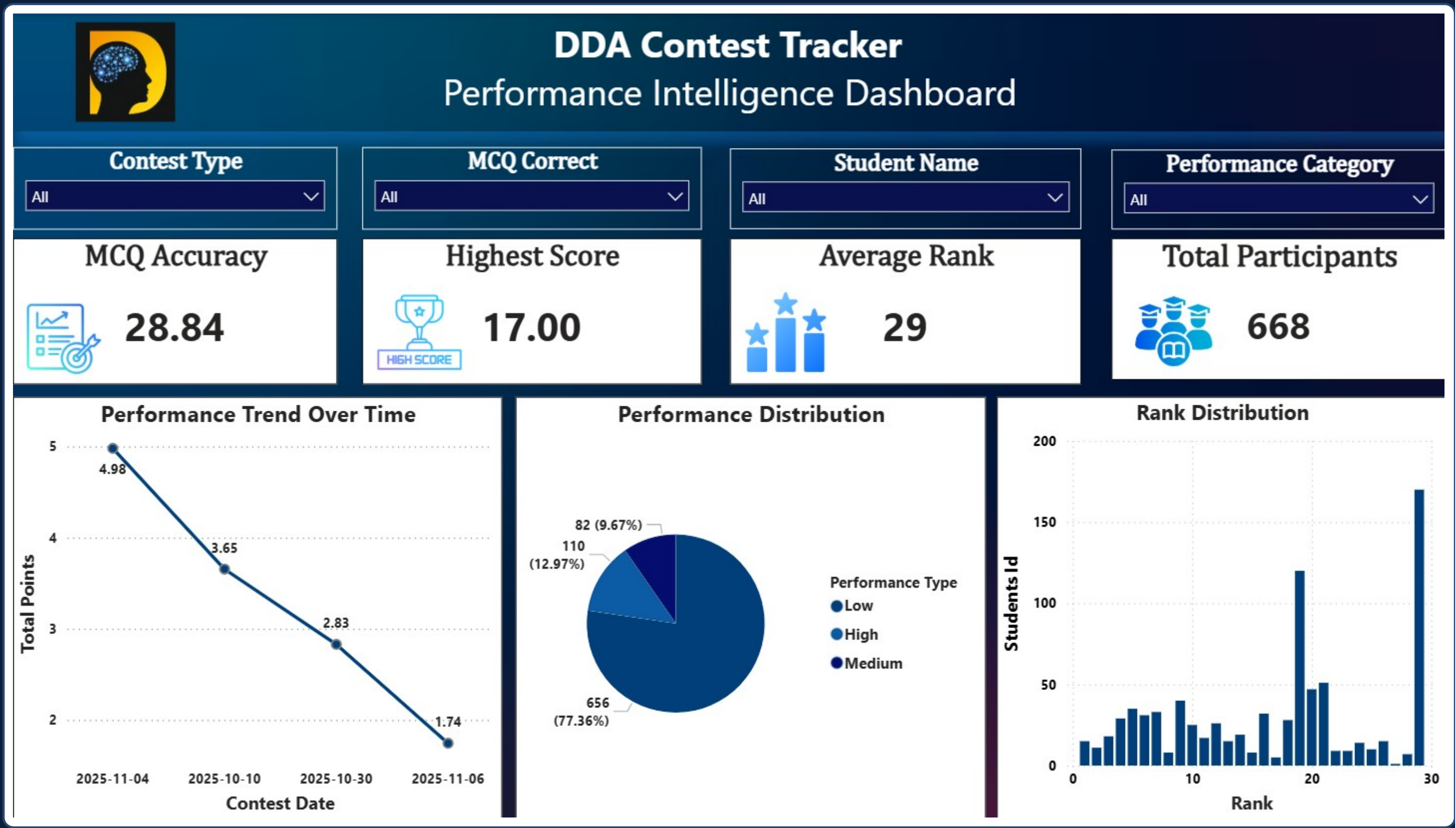
A top-level view of contest activity — total students, branch coverage, average accuracy, and average points — filterable by contest type, date, branch, and rank.

668	Total Students
12	Branches Covered
28.84	Accuracy Score

Also visualizes participation by contest type, participation per branch, and accuracy comparison across contests.

DASHBOARD 02

Student Performance & Competitive Analytics



WHAT IT ANALYZES

A deeper, student-level view — MCQ accuracy, highest score, average rank, performance trend over time, performance-tier distribution, and rank distribution — filterable by contest type, MCQ correct, student name, and performance category.

17.00	Highest Score
29	Average Rank
668	Total Participants

FINDINGS

Key Insights

Overall Performance

Average total points trended downward across the four tracked contest dates — 4.98 → 3.65 → 2.83 → 1.74 — pointing to a gap worth investigating.

Accuracy

Overall accuracy score sits at 28.84, with the highest single score recorded at 17.00 — showing clear room for improvement in precision.

Programming vs. MCQ

The dashboard separately tracks MCQ correctness and programming points, allowing skill-specific coaching rather than one blanket score.

Branch Participation

CSIT, CSE, and AIML lead participation volume, while several branches show minimal engagement — highlighting an outreach gap.

Contest Trends

Accuracy varies widely by contest, from 49.67 on Level 2 Code Clash down to 0.00 on 06 Nov MCA, showing inconsistent difficulty or engagement.

Performance Tiers

Students are grouped into Low, Medium, and High performance categories, with the large majority currently concentrated in a single tier — a key mentoring signal.

NEXT STEPS

Recommendations

Targeted Mentoring

Direct mentoring toward students identified in the low-performance tier.

Programming Skill Development

Run focused practice sessions to lift programming_point and programming_solved scores.

Accuracy Improvement

Address the accuracy gap through timed practice and review of MCQ patterns.

Continuous Performance Tracking

Keep monitoring performance trends across every new contest cycle.

Branch-Level Interventions

Design branch-specific engagement plans where participation is lowest.

Recognize Top Performers

Publicly celebrate high performers to sustain motivation and healthy competition.

TOOLKIT

Tools & Skills

Power BI	SQL / MySQL	Advanced Excel	DAX
Data Modeling	Data Visualization	Data Cleaning	KPI Development
Business Intelligence	Analytical Thinking	Dashboard Design	

END TO END

Project Impact & Analytics Workflow

Raw Data

Data Cleaning

Data Modeling

DAX & KPIs

Power BI Dashboard

Insights

Data-Driven Decisions

Who this helps

Students

See where they stand and what to improve next.

Mentors

Spot who needs support before it becomes a pattern.

Training Teams

Design contests around real skill gaps.

Management

Track program-wide outcomes at a glance.

WRAP UP

Conclusion

The DDA Contest Tracker demonstrates a complete analytics workflow — from four raw source tables, through a clean star-schema model, to two interactive Power BI dashboards. It converts contest and leaderboard data into clear signals about participation, accuracy, and programming performance at both the individual and branch level.

The result is a reusable foundation that mentors and training teams can act on immediately — and a portfolio piece that shows the full data analyst workflow end to end.

> ***Turning student performance data into meaningful insights for better learning outcomes.***

Thank You

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